LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 12 Test (Set B): Magnetic Effects of Electric Current

Chapter / Topic: Magnetic Effects of Electric Current

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.

SECTION A

6 marks

Choose the correct option.

Q1. Outside a bar magnet, magnetic field lines go: **[1]**

(a) from the north pole to the south pole
(b) from the south pole to the north pole
(c) in straight lines away from both poles
(d) only near the middle of the magnet

Q2. A current-carrying conductor placed parallel to a magnetic field experiences: **[1]**

(a) the maximum force (b) a force along the field
(c) a force opposite to the field (d) no force

Q3. Which instrument detects the presence of a current in a circuit? **[1]**

(a) Voltmeter (b) Galvanometer
(c) Rheostat (d) Solenoid

Q4. Which of the following increases the strength of an electromagnet? **[1]**

(a) Reducing the current (b) Using a soft iron core
(c) Reducing the number of turns (d) Using a wooden core

Q5. Which safety device works on the heating effect of current? **[1]**

(a) Earth wire (b) Voltmeter
(c) Solenoid (d) Fuse

Q6. **Assertion (A):** Appliances with metal bodies are connected to the earth wire.
Reason (R): The earth wire carries the main current to the appliance. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Describe the magnetic field lines of a current-carrying circular loop. Why is the field strong at the centre? **[2]**

OR

Why does a compass needle get deflected when it is brought near a bar magnet?

Q8. Give two precautions that help avoid overloading of domestic circuits. **[2]**

SECTION C

6 marks

Q9. A circular loop of wire lies flat on a table, and current flows through it clockwise (as seen from above). **[3]**

(i) State the right-hand thumb rule. **[1]**

(ii) What is the direction of the magnetic field inside the loop? **[1]**

(iii) What is the direction of the magnetic field outside the loop? **[1]**

Q10. Give three ways to increase the force on a current-carrying conductor placed in a magnetic field. **[3]**

OR

A house has a 5 A lighting circuit. Can (i) a 100 W, 220 V bulb and (ii) a 2000 W, 220 V heater be used safely on it? Show your calculation.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Describe an activity to show that a current-carrying conductor experiences a force in a magnetic field. **[3]**

(ii) What happens to the displacement of the rod if both the current and the magnetic field are reversed? Explain. **[2]**

OR

(a) Describe an activity to show the magnetic field around a straight current-carrying wire. [3]
(b) How does this field change if the current is increased, or if we move farther from the wire? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. An electric current always produces a magnetic field. Even weak ion currents travelling along the nerve cells in our body produce magnetic fields. The heart and the brain are the two main organs where these fields are significant. A technique called Magnetic Resonance Imaging (MRI) uses magnetic fields to obtain images of internal organs, which helps doctors in diagnosis. **[4]**

(i) What produces magnetic fields inside the human body? **[1]**

(ii) Name the two organs where these magnetic fields are significant. **[1]**

(iii) What is MRI, and why is it useful? **[2]**

~ All the best! ~